Hong Jun Lee

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EDUCATION

King's College London *Jun 2026 – Present*
PhD in Computer Science *London, United Kingdom*
Advisors: Stefanos Leonardos, Yali Du
Research: Multi-Agent Reinforcement Learning, Meta-Learning, Game Theory

London School of Economics and Political Science (LSE) *Sep 2022 – Jul 2023*
MSc Quantitative Methods for Risk Management *London, United Kingdom*

Heriot-Watt University *Sep 2019 – Jun 2022*
BSc (Hons) Actuarial Science *Edinburgh, United Kingdom*
First Class Honours
Deputy Principal's Award for Academic Excellence (2021/22)

PUBLICATIONS

Discovering Update Rules for Opponent Shaping in General-Sum Games
Hong Jun Lee, Ziyan Wang, Jun Wang, Stefanos Leonardos, Yali Du *Under review, AAMAS 2027*

EXPERIENCE

Tokio Marine Kiln *Feb 2025 – Jun 2025*
Cyber Data & Catastrophe Analyst *London, United Kingdom*
- Validated portfolio data and forecast cyber-loss trends with Bayesian models and logistic regression, and traced the drivers of shifts in modelled losses, including business-interruption losses.
- Benchmarked vendor cyber-catastrophe models against in-house estimates, using Monte Carlo simulation of heavy-tailed (Poisson–Pareto) losses to compare exceedance-probability curves.
- Led the upgrade of the core catastrophe-modelling platform, coordinating IT and InfoSec.

Deloitte UK *Sep 2023 – Nov 2024*
Actuarial Analyst *London, United Kingdom*
- Built a Python (PyQt) tool for running economic scenario generators (interest rates, yield curves, implied volatility), and cut its runtime by 30% by vectorising the Monte Carlo loops in NumPy.
- Modelled yield-curve dynamics with principal component analysis (PCA: level, slope, curvature) and interest-rate volatility surfaces with the SABR stochastic-volatility model.
- Ran economic-capital simulations (99.5% one-year VaR) to quantify the capital impact of methodology changes, and calibrated proxy capital models with simulated annealing and SGD.

London School of Economics and Political Science (LSE) *Mar 2023 – Jul 2023*
Research Assistant *London, United Kingdom*
- Studied major investment banks' risk management through the 2008 crisis, forecasting volatility and correlation with GARCH and DCC models and backtesting Value-at-Risk.
- Contributed to the syllabus of a summer course on Quantitative Methods for Finance and Risk Analysis, covering copulas, time-series models and stress testing.

TEACHING

King's College London *Oct 2026 – Present*
Teaching Assistant *London, United Kingdom*
6CCSARDM Agent Reasoning and Decision Making
7CCSMQMF Quantitative Methods in Finance
7CCSMSCF Scientific Computing for Finance

SKILLS & INTERESTS

Programming: Python (JAX, PyTorch, NumPy)
Methods: Machine Learning, Stochastic Calculus, Statistics, Optimization
Languages: English, Mandarin, Malay, Japanese
Interests: Chess, Badminton, Reading